

David Bulczak

SOFTWARE ENGINEER · GAME DEVELOPER · COMPUTER GRAPHICS EXPERT

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I am a seasoned Game Programmer with a passion for technology and a deep love for the world of gaming. With extensive experience in game development and a strong foundation in software engineering, I excel in solving complex problems and possess a specialization in computer graphics. My analytical prowess, mathematical skills, and dedication to innovation make me a versatile professional who thrives on challenges and consistently delivers high-quality results. Additionally, my leadership skills have been honed through guiding and leading teams to success in the dynamic field of game development and prototyping. Beyond my technical expertise and passion for game development, I am a creative individual who enjoys making music, exploring comics, and maintaining a balanced lifestyle through fitness.

Skills

Programming	C/C++, C#, Python, MATLAB, TypeScript, JavaScript, Java, Lua, SQL, php, Latex
Web & Cloud	React, Next.js, Node.js, GraphQL, Supabase, Firebase, Docker, Kubernetes, AWS, Google Cloud
Frameworks and Engines	Unreal Engine, Unity, VR/AR/XR, Mobile (Android, iOS, Flutter), OpenGL, OpenCL, Vulkan, Shader development
AI	AI-Assisted Development (Claude Code, Copilot), LLM Integration, Prompt Engineering
Others	Microservices, Linux, Windows, Perforce, Git, Project Management, Scrum, Agile, CI/CD, Automated Testing
Languages	German, English, Polish, Japanese

Experience

Aesir Interactive GmbH

SENIOR PROGRAMMER

Remote & Munich, Germany

01.2024 - 09.2026

- Core Systems Architecture: Development of complex gameplay mechanics - including Interaction, Inventory, Crafting and Building systems - heavily leveraging Unreal Engine's Gameplay Ability System (GAS).
- Multiplayer Engineering: Designed and implemented robust network replication across all assigned gameplay features, ensuring strict multiplayer readiness and seamless synchronization.
- Data Pipelines & Tooling: Engineered a custom Python-based import/export data pipeline, empowering game designers to efficiently iterate, balance, and integrate gameplay data directly via Excel.
- Marketing Integration: Developed specialized in-game capturing toolsets, providing the marketing department with an optimized workflow for generating high-quality promotional assets.
- Cross-Disciplinary Collaboration: Bridged the gap between engineering, design, and marketing by delivering tailored, user-friendly tools that significantly improved inter-departmental workflows.

Farbspiel Interactive GmbH & Co. KG

LEAD PROGRAMMER & PRODUCER

Cologne, Germany

05.2021 - 12.2023

- Designed and built the studio's technical toolset, stack and infrastructure, including CI/CD pipelines with automated testing
- Established code review processes and quality assurance standards
- Designed and implemented modularized, scalable gameplay and engine systems in Unreal Engine (Blueprint & C++), with a focus on performance
- Migrated the project from Unreal Engine 4 to Unreal Engine 5 during active production
- Managed the engineering team
- Took on production/project management from Jan. 2023, leading it from Jul. 2023 to ensure on-time delivery

Macromedia University of Applied Science

LECTURER

Cologne, Germany & remote

10.2021 - 06.2025

- Organization and deliver lectures for "Game Engines" and "Software Project"
- Project management for student projects in "Game Engines" lecture

Rivers and Wine Studio GmbH

GAMEPLAY PROGRAMMER

Cologne, Germany

02.2020 - 06.2021

- Designed and implemented gameplay systems in Unreal Engine (Blueprint & C++)
- Modularized the existing codebase and introduced coding standards and code review processes
- Built a CI/CD pipeline with an automated test suite
- Optimized tech-art related aspects of the game

aifora GmbH

BACKEND SOFTWARE ENGINEER

Dusseldorf, Germany & remote

07.2020 - 12.2020

- Developed and extended services in aifora's microservice-based retail automation platform, working within their engineering team
- Implemented robust, scalable backend components

Extality

SOFTWARE ENGINEER

Cologne, Germany

11.2018 - 02.2020

- End-to-end development and delivery of multiple VR/AR/MR client applications in Unity and Unreal Engine, shipped to Android and iOS
- Built shared multi-user AR, synchronizing several mobile devices into one co-located AR space
- Led project management and software prototyping for internal projects
- Modularized and maintained the software stack, and established coding conventions

Computer Graphics and Multimedia Systems Group, University of Siegen

Siegen, Germany

RESEARCH ASSISTANT

08.2014 - 10.2018

- Researched and developed physically-based simulation methods for Time-of-Flight sensors in collaboration with an industrial partner, publishing results in peer-reviewed papers
- Built scenarios, scenes and datasets for quantitative evaluation of Time-of-Flight algorithms
- Built Python and MATLAB pipelines for sensor data processing, analysis and evaluation, and trained neural networks with TensorFlow
- Taught Computer Graphics lectures, tutorials and practical training courses
- Supervised student research assistants

LOKISA Smart Energy GmbH

Meckernich, Germany

SOFTWARE ENGINEER

01.2013 - 08.2014

- Developed a smart energy manager for photovoltaic systems, including low-level C++ control software on Raspberry Pi and Pironex piA single-board computers
- Implemented the Modbus protocol over RS-232 and RS-485 in C/C++, and evaluated and integrated the Solutronic protocol
- Built a server backend and GUI for the platform, and an internal sales CRM in Delphi
- Customized the underlying Linux system and maintained supporting libraries
- Built 3D rendering tools for internal planning purposes, working in a Scrum team

Education

RWTH Aachen University

Aachen, Germany

DIPL.-INFORM. IN COMPUTER SCIENCE (DIPLOMA)

10.2005 - 10.2012

- Major subject: Computer Graphics and Multimedia
- Minor subject: Mathematics
- Title of thesis: Computing Data Driven Facial Animations from Sparse Keyframes
- Thesis mark: very good
- Overall mark: good

Additional Experience

Gravinator Games

remote

UNREAL ENGINE ENGINEER

03.2018 - 12.2020

- Hobby project: prototyped a multiplayer hero-based RTS/economic simulation game in Unreal Engine (Blueprint and C++)

Computer Graphics Group, RWTH Aachen University

Aachen, Germany

STUDENT RESEARCH ASSISTANT

10.2012 - 12.2012

- Creating and maintaining database with facial animations
- Implementation of improvements for facial animation tools

Publications

Efficient Subsurface Scattering for Time-of-Flight Sensors

EG VMV 2018

[HTTPS://DIQLIB.EG.ORG/HANDLE/10.2312/vmv20181259](https://diqlib.eg.org/handle/10.2312/vmv20181259)

2018

- Also presented as a talk at GCPR VMV Conference, 2018

Quantified, Interactive Simulation of AMCW ToF Camera Including Multipath Effects

MDPI Sensors 2018, 18(1)

[HTTPS://WWW.NCBI.NLM.NIH.GOV/PMC/ARTICLES/PMC5796323/](https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5796323/)

2018

Fast Motion Estimation for Field Sequential Imaging: Survey and Benchmark

Image and Vision Computing 89,

2019

[HTTPS://WWW.SCIENCEDIRECT.COM/SCIENCE/ARTICLE/PII/S0262885619301003](https://www.sciencedirect.com/science/article/pii/S0262885619301003)

2019

Interests & Hobbies

- Spending time with my animals
- Workout
- Making music in my studio
- Comics
- Videogames